Robust Multimodal Remote Sensing Image Matching Using Edge Consistency Scale-Space and Significant Relative Response

Zhonghua Hong, *Member, IEEE*, Jinyang Chen, Xiaohua Tong, *Fellow, IEEE*, Shijie Liu, *Member, IEEE*, Ruyan Zhou, Haiyan Pan, and Qing Fu

***Abstract*— Multimodal remote sensing images (MRSIs) often suffer from severe nonlinear radiation distortions (NRDs) and significant geometric distortions, making precise matching challenging. We developed a feature-based matching algorithm to address this issue using edge consistency scale-space (ECSS) and significant relative response. By designing an edge consistency filtering (ECF), we construct a scale space that preserves the structural information of MRSI at various scales, enhancing scale invariance. ECSS computes feature descriptors using multiorientation filtering techniques to construct significant relative responses. This approach not only resists NRD but also utilizes information from all directional filters to build descriptors with higher discriminative power compared to direct filter responses or the maximum index map (MIM). To ensure rotational invariance, ECSS employs a robust technique for estimating the primary orientation. To further optimize matching accuracy and increase the number of effective matching points, ECSS uses a coarse-to-fine matching strategy. This involves using preliminary matching results to estimate the affine transformation between images, which then guides a more refined secondary matching. We evaluated the performance of ECSS on five different MRSI datasets and compared the results with nine state-of-the-art matching methods: ReDFeat, MINIMA-LG, radiation–variation insensitive feature transformation (RIFT), MS-HLMO, SRIF, weighted structure saliency feature (WSSF), POS-GIFT, oriented filter-based matching (OFM), and GLS-MIFT. The experimental results demonstrate that ECSS excels in all performance metrics, particularly in terms of stability and matching accuracy when handling MRSI data with high NRD and complex geometric transformations.**

***Index Terms*— Feature matching, multimodal image matching, nonlinear radiation distortion (NRD), phase congruency (PC).**

I. Introduction

IMAGE matching technology plays a pivotal role in essential applications, such as bundle block adjustment [1], [2], ortho-rectification [3], image mosaic [4], and 3-D reconstruction [5], [6]. This technology is also widely used in fields like SLAM [7], image registration [8], image fusion [9], [10], disaster detection [11], and navigation [12], [13]. With the rapid development of multisource sensor technology, image data from different angles, times, and sensors are extensively used to capture detailed information about the terrestrial landscape, enriching image content. Multimodal remote sensing image (MRSI) matching technology, which involves matching remote sensing images from different types of sensors, has become a hot topic in current research. Nevertheless, MRSI matching still faces several inherent challenges. First, different sensors may capture images at different resolutions, requiring scale conversion and adjustment. Second, due to differences in sensor installation positions and imaging angles, there may be rotational deviations between images, necessitating correction. More complexly, unlike matching homologous images, multimodal images exhibit nonlinear radiation distortion (NRD), which arises from different imaging mechanisms of different sensors, causing the same objects to display different radiometric characteristics in different modal images. Despite decades of development in image-matching technology [14], [15], the challenge of matching images across multiple modes continues to be a significant hurdle due to these issues.

In recent decades, the classification of image matching algorithms has typically encompassed three main types: area-based methods, feature-based methods, and deep learning-based methods [16]. Normalized cross correlation (NCC) [17] and mutual information (MI) [18] are the most common area-based methods, using calculations of intensity or MI between image blocks to evaluate and match based on similarity. The advantage of this method is its high matching accuracy [19], [20], [21]; however, it is usually sensitive to scale and rotational differences between images and heavily relies on prior information about the images. Feature-based matching methods typically start by identifying salient feature points in an image and constructing descriptors for these points. Then, these methods establish correspondences by finding the most similar pairs of descriptors between two images. These methods are robust to rotational and scale variations between images. Traditional feature-based matching methods, such as scale-invariant feature transform (SIFT) [22], speed-up robust features (SURFs) [23], and KAZE [24], primarily rely on image gradients to characterize feature points. Yet,

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these gradient descriptors are sensitive to NRD, resulting in these traditional methods often failing to achieve good matching results in MRSIs. Methods based on deep learning have achieved considerable advancements within the realm of computer vision, with approaches like Superglue [25] and LoFTR [26] demonstrating considerable potential. However, these methods heavily rely on extensive data collection; their matching performance may drastically decline when faced with previously untrained data. Given the diversity and complexity of MRSI data, deep learning-based methods are not yet the optimal solution.

Recent studies have shown that approaches based on multiorientation filter information, instead of traditional gradient information, have achieved good results in MRSI. Such approaches build local descriptors using the filter's directional characteristics, effectively resisting NRD, but they cannot achieve rotational invariance through traditional methods of calculating the primary orientation of image gradients. For example, radiation–variation insensitive feature transformation (RIFT) [27] achieves rotational invariance by constructing convolution sequence rings, but this method is too time-consuming to simultaneously achieve scale invariance. Position-orientation-scale guided geometric and intensity-invariant feature transformation (POS-GIFT) [28] attempts to determine the primary orientation by summing the filter results within subregions of the descriptor from different directions, but this reliance on descriptor structure makes its rotational invariance unstable. In contrast, oriented filter-based matching (OFM) [29] estimates the primary orientation by computing the centroid of the descriptor's most prominent region, while modality-invariant feature transform with global-to-local searching (GLS-MIFT) [30] uses the directional characteristics of the multiorientation filter itself to compute the primary orientation, both achieving good rotational invariance. However, like RIFT, both methods use the maximum index to construct the descriptor, which, while resistant to NRD, does not fully utilize all the information from the multiorientation filter. Additionally, they use Gaussian filtering (GF) to build image pyramids, which may lead to the loss of too many structural features in tasks involving matching images with significant scale differences and indistinct structural features, resulting in matching failures. Yao et al. [31] proposed co-occurrence filtering (COF) to construct scale space, aiming to preserve structural information in images and extract more edge features. However, when dealing with structurally complex images, this filter may cause edge blurring, reducing edge sharpness. Weighted structure saliency feature (WSSF) [32] uses pointwise shape-adaptive texture filtering (PSATF) to construct its scale space. This technique excels at preserving prominent structures but performs less effectively on images with weaker features.

In this article, a novel method is devised for matching MRSI based on edge consistency scale-space and significant relative response (ECSS). We innovatively introduced the edge consistency filter (ECF), which, while smoothing images, effectively preserves critical edge information crucial for maintaining the structure of the image. Utilizing the ECF, we constructed an edge consistency scale space that maintains the structural information of images across various scales, thereby achieving exceptional scale invariance. Furthermore, we calculated significant relative responses over multiorientation, multiscale filter results, creating descriptors that effectively combat NRD. These descriptors not only utilize the directional nature of multiorientation filtering to estimate a robust primary orientation but also maximize the use of all information from the multiorientation filters. Finally, we employed an affine transformation-based rematching strategy to further refine the matching results, significantly enhancing the accuracy and reliability of the matches. The key contributions of this article are outlined as follows.

1) We developed a novel feature matching algorithm named ECSS. The algorithm effectively achieves feature matching for multimodal remote sensing images with significant scale and rotational variations. Extensive experiments on diverse datasets with substantial scale and rotational differences demonstrate that ECSS outperforms nine state-of-the-art multimodal matching methods, including two deep learning-based methods, ReDFeat, MINIMA-LG, and seven handcrafted-based methods, such as RIFT, MS-HLMO, SRIF, WSSF, POS-GIFT, OFM, and GLS-MIFT.
2) We designed a novel edge-preserving filter, ECF, which effectively retains edge information while smoothing images. This characteristic better preserves the structural information of images when constructing scale spaces. Employing this method enables the detection of more and more accurate feature points, thereby enhancing the overall matching performance.
3) We developed a method to construct descriptors using significant relative responses from multiorientation filters, which possess good stability and effectively resist NRD. Additionally, we employed a robust method for estimating the principal orientation, ensuring that descriptors maintain rotational invariance across the full range of [0°, 360°].
4) We implemented a rematching strategy based on affine transformations aimed at optimizing initial matching results. This strategy considerably improves the accuracy and increases the count of matching points through optimization and refinement of the matching process.

II. Related Work

Currently, the matching algorithms for MRSIs are primarily classified into three categories: area-based methods, feature-based methods, and deep learning-based methods.

A. Area-Based Methods

Area-based methods typically involve establishing a template window and sliding it over the image to calculate the similarity of matching in different areas. The key to this method lies in establishing an effective measure of similarity. Traditional area-based methods, such as NCC [17], use image intensity information as a measure of similarity and can effectively handle linear intensity variations in images. Hence, they are often used for remote sensing image matching [33], [34].

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However, NCC is susceptible to NRD, making it unsuitable for MRSI matching tasks [35]. Another common method of similarity measurement is MI [18], [36], which is somewhat resistant to NRD [37], [38], but prone to local optima and high computational demands, limiting its applicability. Recently, several customized area-based methods have been developed for multimodal image matching. As an illustration, Ye et al. [39] combined the histogram of oriented gradients (HOGs) [40] with phase congruency (PC) to propose the histogram of orientated PC (HOPC) method, which utilizes structural information within images for matching, enabling it to handle multimodal images. Fan et al. [21] enhanced matching results by combining PC with nonlinear diffusion techniques. To achieve rapid matching of MRSI, Ye et al. [41] used the fast Fourier transform to propose a dense structured image template method named channel feature of oriented gradients (CFOGs). Xiang et al. [42] introduced an enhanced PC model that achieves superior outcomes in optical-SAR image matching. Ye et al. [43] combined first- and second-order gradient information using multiscale steerable filtering to construct a structural descriptor called the steerable filters of first- and second-order channel (SFOC), which effectively resists NRD. Although area-based matching methods can attain high accuracy, they are typically suited only for images undergoing translational transformations, which constrains their effectiveness in MRSI matching.

B. Feature-Based Methods

Feature-based methods have rapidly advanced over the past several years. These methods typically identify stable and prominent features in images, construct descriptors for each feature, and match them using specific evaluation techniques. Classic matching methods like SIFT [22], SURF [23], and affine-SIFT (ASIFT) [44] have been widely applied across various fields. To meet the needs for real-time matching, oriented fast and rotated brief (ORB) [45] uses binary descriptors and Hamming distance for rapid matching. However, due to severe multiplicative noise issues in SAR images, traditional intensity-based algorithms perform poorly on such images, leading to the introduction of ROEWA filtering in SAR-SIFT [46] for matching. Hong et al. [47] further enhanced the accuracy of SAR-SIFT by using an unbiased difference ratio edge detector. Despite years of improvements, most of these algorithms are still limited to homogenous image matching. Xiang et al. [48] extended the applicability to optical and SAR image matching by combining Sobel filtering with ROEWA filtering. Position-scale-orientation SIFT (PSO-SIFT) [49] employs second-order gradients for descriptor construction and estimates the optimal geometric transformation model to enhance resistance to NRD and demonstrate its matching capabilities on optical-infrared images. Aguilera et al. [50] proposed the LGHD algorithm that utilizes multiscale, multiorientation log-Gabor filters, offering excellent performance in multimodal image matching, yet it struggles with images exhibiting rotational and scale variations. RIFT [27] discards filter response values and instead uses the maximum response direction index from various orientation filters to construct a maximum index map (MIM), which serves to develop feature descriptors and achieve rotational invariance with convolution sequence rings. This approach has proven more resistant to NRD and has achieved good matching performance in MRSI, but the method for achieving rotational invariance in RIFT is too time-consuming and lacks scale invariance. Xiong et al. [51] introduced the oriented self-similarity (OSS) method, leveraging self-similarity features to accomplish multimodal image matching. Zhang et al. [52] developed the multidimensional OSS (MOSS) features method based on OSS. Locally normalized image feature transform (LNIFT) [53] uses locally normalized images as intermediate state images to resist NRD and unifies the direction to [0°, 180°) during descriptor construction to avoid gradient direction reversal. Yao et al. [31] reduced the impact of NRD by combining COF with second-order gradient features. Multiscale histogram of local main orientation (MS-HLMO) [54] obtains descriptors resistant to NRD through the generalized gradient location and orientation histogram (GGLOH). Li et al. [55] use local intensity binary transformation (LIBT) as an intermediate image for feature detection and description, achieving good matching results. OFM [29] builds upon RIFT by evaluating multiscale, multiorientation filters, calculating the primary orientation using the centroid of the descriptor's dominant region, and establishing a Gaussian scale space to achieve scale invariance. Wan et al. [32] proposed WSSFs by combining phase information and machine learning, better utilizing image structural information to resist NRD. POS-GIFT [28] achieves numerous matching points on most images, but the direct use of log-Gabor filter amplitudes to construct descriptors performs poorly on images with significant NRD. GLS-MIFT [30] has created unique feature point detection methods and designed adaptive partition descriptors, achieving excellent performance in multimodal image matching.

C. Deep Learning-Based Methods

In recent years, deep learning-based methods have gained significant attention due to their potential to revolutionize various fields. SuperPoint [56] utilizes a self-supervised approach to train a feature detector and descriptor generator, demonstrating performance that surpasses traditional feature detection techniques on its dataset. SuperGlue [25] employs self-attention and cross-attention mechanisms to learn feature points and descriptors, and in conjunction with SuperPoint, it exhibits robust matching capabilities. LoFTR [26] adopts a semi-dense matching strategy, discarding traditional feature detection steps, and extracts and matches image descriptors through self-attention and cross-attention layers, showing significant matching advantages in feature-sparse areas. LightGlue [57] can automatically assess the difficulty of image matching and adaptively adjust the matching mode, thus enhancing matching efficiency while ensuring results. FeMIP [58] features a system with coarse matching and fine regression modules, achieving significant success in multimodal image matching. Quan et al. [59] created SDNet, a network tailored specifically for optical-SAR image matching, which improves matching performance through

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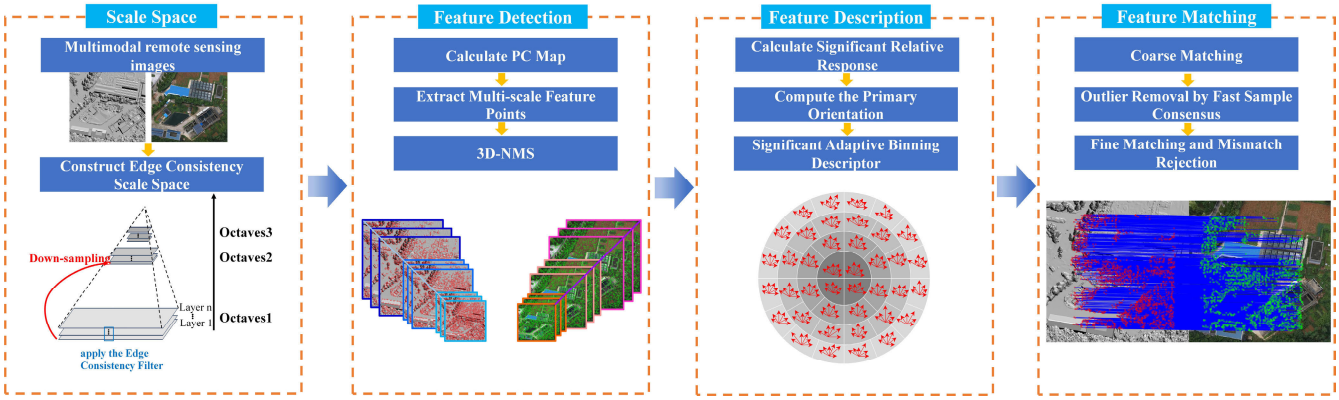

Fig. 1. Flowchart of the ECSS.

enhancements in network structure and optimization. Ye et al. [60] developed MU-Net, a multiscale unsupervised learning framework that can directly learn the end-to-end mapping of transformation parameters from image pairs, demonstrating strong matching performance on MRSI. Deng and Ma [61] employed a mutual weighting strategy to re-couple the independent constraints of detection and description in multimodal feature learning, improving both training stability and the performance of learned cross-modal features. MINIMA [62] uses a simple data engine to freely generate large datasets with multiple modalities, diverse scenes, and accurate matching labels, achieving excellent results on multimodal images. To further improve matching accuracy in area-based matching methods, Ye et al. [63] proposed a hybrid matching method based on attention-enhanced structural features (AESFs), combining both handcrafted and learned features. By utilizing AESF, more prominent image structural features can be obtained. Despite the immense potential of deep learning-based methods in image matching, these techniques highly depend on training datasets in practical applications, most of them often performing poorly with untrained new images, lacking generalization capabilities [55].

As summarized from previous research, although some progress has been made, severe geometric differences and NRD in MRSI continue to pose significant challenges in image processing. Area-based methods, although advantageous in accuracy, cannot handle scale and rotational variations and typically rely on prior geographic information from MRSI for preliminary matching, limiting their versatility in images without prior information. Deep learning-based methods, while showing potential, are heavily dependent on specific training datasets, limiting their applicability. Feature-based methods exhibit better versatility, yet most existing methods still fall short in simultaneously resisting NRD and addressing geometric differences between images. To address these challenges, we introduce a new feature-based matching algorithm named ECSS. ECSS utilizes an edge-preserving filter to maintain the structural features of images and constructs a scale space accordingly to achieve scale invariance. It constructs descriptors using multiscale, multiorientation filters and employs a robust strategy for estimating the principal orientation to achieve rotational invariance. Additionally, through a coarse-to-fine matching approach, ECSS significantly improves the accuracy and number of correct correspondences. Combining these features, ECSS effectively captures substantial correspondences in images with significant geometric differences, demonstrating considerable advantages in complex MRSI matching tasks.

III. Methodology

As shown in Fig. 1, the ECSS algorithm's processing flow includes four key steps. In the initial stage, scale space is constructed through edge consistency filtering to ensure scale invariance. Subsequently, a substantial number of corner points and edge points are detected on both the maximum moment and the minimum moment using the FAST feature detector [64]. In the third step, feature descriptors for each feature point are calculated using the significant relative response, and the descriptors are adjusted according to the primary orientation to generate feature description vectors with rotational invariance. In the final stage, correspondences are initially obtained using brute-force matching and the FSC algorithm [65]. These correspondences are leveraged to derive an affine transformation matrix between the two images. This matrix is applied to perform an affine transformation on the images to be matched, eliminating rotational and size differences with the reference image. After transformation, feature descriptors are recalculated and matched with the descriptors of the reference image. This progressive, coarse-to-fine matching process optimizes the match results.

A. Edge Consistency Scale Space

Constructing scale space is a critical component of feature matching algorithms. Due to NRD in MRSI, traditional isotropic and anisotropic filtering methods often struggle to effectively preserve the edge structures of images; however, edge structures play a vital role in multimodal image matching. To tackle this issue, we developed a new type of filter called the edge consistency filter (ECF), which more effectively preserves the edge structures of images while smoothing them.

1) Edge Consistency Filter: ECF is an edge-preserving filter based on joint bilateral filtering [66], with its definition

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shown as

$$J_p = \frac{\sum_{q\in N(p)} G_{\sigma_s}(p,q)\cdot M\left(I_p, I_q\right)\cdot I_q}{\sum_{q\in N(p)} G_{\sigma_s}(p,q)\cdot M\left(I_p, I_q\right)}. \tag{1}$$

J_p and I_q are the output pixel and the input pixel; p and q are the positions of the pixel indexes in the image; $G_{\sigma_s}(p,q)\cdot M(p,q)$ is the weight contributed by pixel q to output pixel p; $G_{\sigma_s}(p,q)$ is a Gaussian filter with a standard deviation of σ_s; $M(I_p, I_q)$ is the guide image to determine whether I_p and I_q are on the same side of an edge.

2) Edge Consistency Guide Image: Computing guide images is a critical aspect of using the ECF. To effectively preserve image edge information, it is essential to ensure that only pixels on the same side of an edge as the current pixel participate in the filtering process. This approach prevents pixels on opposite sides of an edge from influencing each other, thereby maintaining edge clarity and consistency. The even component of the log-Gabor filter is highly suited for image edge segmentation tasks. In image processing, it generates a positive response at edges from low to high pixel value regions and a negative response at edges from high to low pixel value regions. These response characteristics are highly beneficial for identifying and separating the sides of an edge within an image, thereby distinctly delineating different image regions. By accumulating multiscale, multiorientation filtering results, a more comprehensive image analysis is provided. The process for computing the guide image M is shown as follows:

$$e_{\mathrm{so}}(x,y) = I(x,y) * L^{\mathrm{even}}(x,y,s,o) \tag{2}$$

$$H(d) = \begin{cases} 1, & d > 0 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

$$\begin{cases} M\left(I\left(x_p, y_p\right), I\left(x_q, y_q\right)\right) = H(\alpha) \\ \alpha = \sum_s \sum_o e_{\mathrm{so}}\left(x_p, y_p\right)\cdot \sum_s \sum_o e_{\mathrm{so}}\left(x_q, y_q\right). \end{cases} \tag{4}$$

$e_{\mathrm{so}}(x,y)$ is the response of $L^{\mathrm{even}}(x,y,s,o)$ at scale s and orientation o. In this study, we assigned values of 4 to s and 6 to o, respectively; (x,y) represents the coordinates of an image point; $H(d)$ acts as a binary indicator, which is 1 if d is positive, and 0 otherwise; $M(I(x_p,y_p), I(x_q,y_q))$ represents whether the pixels at positions (x_p, y_p) and (x_q, y_q) are on the same side of the edge.

In fact, the ECF proposed in this article, along with the existing COF and pointwise shape-adaptive texture filter (PSATF), is based on joint bilateral filtering technology. The primary distinction among these methods lies in the computation of the guide image. Specifically, COF assesses the similarity between textures by analyzing the co-occurrence probabilities between two pixels and constructing a 256 × 256 matrix to be used for weighting during the filtering process. Although this method can handle a variety of image textures, it tends to cause blurring in the filtering results when dealing with structurally complex images. PSATF employs structured random forests to calculate the edge confidence map of images, effectively capturing and enhancing edge information, thus optimizing the preservation of structural features during image smoothing. The efficacy of this method significantly depends on the structured random forests' accurate recognition of edges. In our experiments, we noted that PSATF sometimes overlooks slender and elongated edges during the computation of the guidance map, leading to objects on either side of these edges being erroneously identified as the same structure. Such misjudgments can lead to the loss of structural information, consequently affecting the number and accuracy of detected feature points. Fig. 2 displays the results of the three filtering methods; both COF and PSATF exhibit varying degrees of edge blurring, whereas ECF manages to blur unwanted details and noise while preserving structural information.

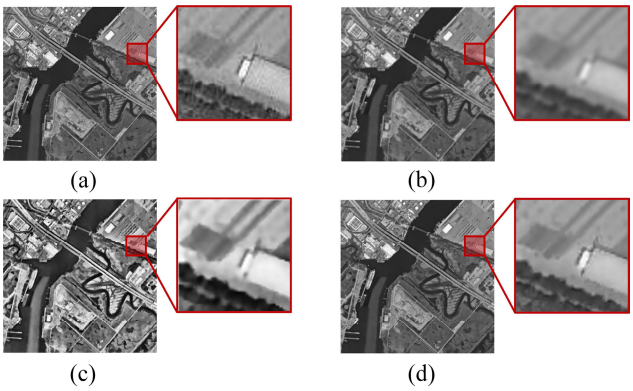

Fig. 2. Filtering results of COF, PSATF, and ECF. (a) Original image. (b) Result of COF. (c) Result of PSATF. (d) Result of ECF.

3) Scale Space Pyramid: Fig. 3 displays the structure of the edge consistency scale space pyramid. The entire pyramid is composed of N octaves. Each octave contains L layers. In the initial step, the original image is subjected to Gaussian blurring to form the first layer of the first octave. Subsequently, filtering with ECF using different standard deviations is performed to generate each layer within an octave, with the image size of each layer in the current octave reduced by a certain proportion compared to the previous octave. The standard deviation for each layer is defined is shown as

$$\sigma_s(n,l) = \sigma_0 \cdot r^{(n-1)+\frac{l-1}{L}}, \quad n \in [1,N], \quad l \in [1,L]. \tag{5}$$

$\sigma_s(n,l)$ denotes σ_s in (1); n and l are the current octave and layer; N and L are the total number of octaves and layers; σ_0 is the initial standard deviation; r represents the scale factor for down-sampling between consecutive octaves in the image pyramid. We set r at 1.5 in this article.

B. Feature Detection

Previous studies have shown that using the FAST feature detector on both the maximum moment and the minimum moment to detect corner and edge points can effectively achieve high repeatability of feature points in image pairs affected by NRD. Therefore, we continue this practice and apply a 3-D nonmaximum suppression (3D-NMS) strategy to each octave to prevent excessive clustering of feature points.

1) Edge Information and Corner Information: The maximum moment efficiently captures the edge information, whereas the minimum moment predominantly reveals the

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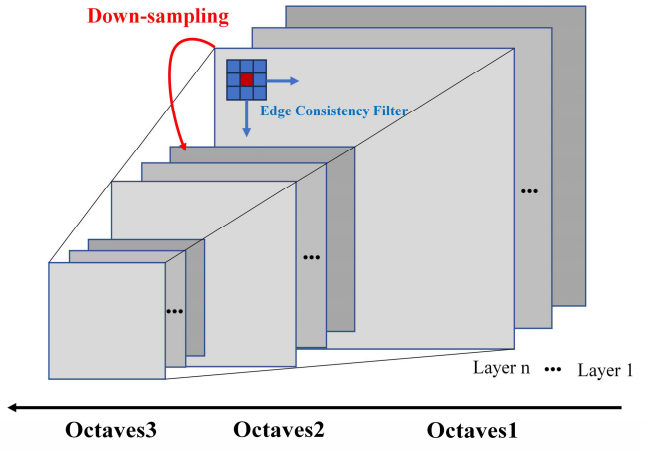

Fig. 3. Structure of the edge consistency scale space.

corner information. To derive the maximum and minimum moment maps, we initially compute phase consistency within each image in the ECF scale space

$$\mathrm{PC}(x,y)=\frac{\sum_s\sum_o \omega_o(x,y)\lfloor A_{\mathrm{so}}(x,y)\Delta\Phi_{\mathrm{so}}(x,y)-T\rfloor}{\sum_s\sum_o A_{\mathrm{so}}(x,y)+\xi}. \tag{6}$$

$\mathrm{PC}(x,y)$ denotes the magnitude of the local PC; (x,y) represents the pixel coordinates; $A_{\mathrm{so}}(x,y)$ denotes the amplitude of the log-Gabor filter at scale s and orientation o; $\omega_o(x,y)$ is a weighting function; $\Delta\Phi_{\mathrm{so}}(x,y)$ denotes a phase deviation; T denotes a noise threshold; $\lfloor\cdot\rfloor$ indicates that the enclosed quantity retains its value when positive or is zero otherwise; ξ represents a small value introduced to prevent the denominator from reaching zero. The processes of calculating PC and constructing scale space can be conducted concurrently to avoid redundant computations of log-Gabor filtering.

Following the calculation of PC, the next step involves calculating the three intermediate quantities essential for generating the maximum moment and minimum moment:

$$\begin{cases} a=\sum_o(\mathrm{PC}(\theta_o)\cos(\theta_o))^2 \\ b=2\sum_o(\mathrm{PC}(\theta_o)\cos(\theta_o)(\mathrm{PC}(\theta_o)\sin(\theta_o))) \\ c=\sum_o(\mathrm{PC}(\theta_o)\sin(\theta_o))^2. \end{cases} \tag{7}$$

$\mathrm{PC}(\theta_o)$ is an independent PC map in the orientation o and θ_o is the angle of orientation o. After that, we can calculate the maximum moment $M_{\max}$ and the minimum moment $M_{\min}$

$$\begin{cases} M_{\max}=\frac{1}{2}\left(c+a+\sqrt{b^2+(a-c)^2}\right) \\ M_{\min}=\frac{1}{2}\left(c+a-\sqrt{b^2+(a-c)^2}\right). \end{cases} \tag{8}$$

$M_{\max}$ represents the maximum moment, which captures the edge information in the image; $M_{\min}$ is the minimum moment, highlighting the corner information within the image. Fig. 4 illustrates the maximum moment and minimum moment of an image.

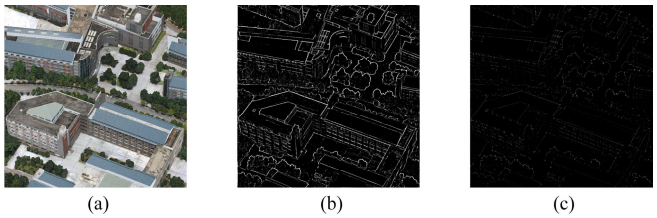

Fig. 4. Establish the maximum moment and minimum moment of an image. (a) Original image. (b) Maximum moment. (c) Minimum moment.

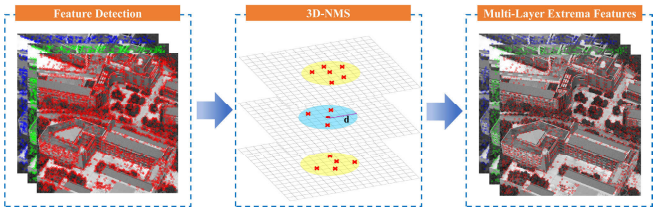

Fig. 5. Feature point detection and 3D-NMS.

2) 3-D Nonmaximum Suppression: Once the maximum moment and minimum moment are obtained, the FAST feature detector is used to identify edge and corner points on each of them. To prevent excessive clustering of feature points, 3D-NMS is applied to each detected feature point. Fig. 5 illustrates the steps involved in feature detection and 3D-NMS. The specific operation of 3D-NMS involves projecting each feature point across all layers within the same octave and calculating the response values of all feature points within a certain radius. A feature point is retained only if its response value is the highest across all layers; otherwise, it is suppressed. The detailed computation approach for 3D-NMS is as follows:

$$\begin{cases} \mathrm{FP}_{\mathrm{nl}}=\left\{\mathrm{kp}_{\mathrm{nl}}(i)|\mathrm{mask}\left(\mathrm{kp}_{\mathrm{nl}}(i)\right)=1\right\} \\ \mathrm{mask}\left(\mathrm{kp}_{\mathrm{nl}}(i)\right)=\prod_{l'=1}^{L}\mathrm{NMS}\left(\mathrm{kp}_{\mathrm{nl}}(i),d,l'\right) \end{cases} \tag{9}$$

$\mathrm{FP}_{\mathrm{nl}}$ is the set of feature points; n and l denote the indexes of octave and layer; $\mathrm{mask}(\mathrm{kp}_{\mathrm{nl}}(i))$ represents whether the feature point is retained or not, 1 means retained and 0 means not retained; $\mathrm{kp}_{\mathrm{nl}}(i)$ represents the feature point at index i located on the lth layer of the nth octave; $\mathrm{NMS}(\mathrm{kp}_{\mathrm{nl}}(i)dl')$ represents whether $\mathrm{kp}_{\mathrm{nl}}(i)$ is the maximum value within radius d at layer l'.

C. Feature Description

Once the feature points of an image are obtained, the subsequent step involves describing each feature point. Traditional feature descriptors are generally constructed based on gradient histograms, but in multimodal images with NRD, gradients tend to be unstable. According to recent research, using multiorientation filters to construct feature descriptors has shown significant effectiveness in multimodal image matching. The log-Gabor filter is employed to compute the significant relative response, generating descriptors with resistance to NRD and further calculating the primary orientation of the descriptors to ensure rotational invariance.

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1) Significant Relative Response: First, we accumulate the filter results across all scales in each orientation to obtain the filter responses for each pixel in various orientations. The specific method for calculating the filter responses is shown as follows:

$$\mathrm{GR}_o(x, y) = \sum_s A_{\mathrm{so}}(x, y). \tag{10}$$

$\mathrm{GR}_o(x, y)$ represents the cumulative response of the log-Gabor filters across all scales in orientation o at position (x, y). $A_{\mathrm{so}}(x, y)$ has been obtained when calculating the phase congruence.

After obtaining the Gabor filter response (GR), we first calculate the mean GR for each pixel across all orientations. Then, we compute the ratio of the GR for each orientation to its corresponding mean value for each pixel

$$\begin{cases} \mathrm{SR}_o(x, y) = \dfrac{\mathrm{GR}_o(x, y)}{\mathrm{AR}(x, y)} \\ \mathrm{AR}(x, y) = \dfrac{\sum_o \mathrm{GR}_o(x, y)}{O}. \end{cases} \tag{11}$$

$\mathrm{SR}_o(x, y)$ represents the significant relative response in the orientation o; $\mathrm{AR}(x, y)$ denotes the mean value of $\mathrm{GR}_o(x, y)$ at position (x, y); O represents the total number of orientations.

Using SR instead of GR for descriptor construction is crucial because, in multimodal images, even the same objects may have different filtering responses in the same orientation. SR emphasizes the relative importance of response values rather than their absolute magnitude, mitigating the influence of local extrema on descriptors. SR also helps reduce the impact of noise by suppressing insignificant variations and highlighting truly significant directional differences.

2) Significant Adaptive Binning Descriptor: After obtaining the SR, we use it to construct descriptors. Based on the AB-SIFT [67] descriptor structure, we designed a circular logarithmic polar coordinate grid with variable partitions. The region to be described is segmented into four distinct rings, and each ring is further subdivided into 4, 8, 12, and 16 subregions, respectively. Subsequently, a directional histogram is created for each subregion. Considering that pixels closer to the center point are usually more relevant to the feature point, to reduce the influence of more distant pixels on the descriptor, we apply Gaussian weighting to the descriptor. Specifically, the SR of each pixel in all orientations is multiplied by a weight, and the results are added to the corresponding histogram. Fig. 6 shows the process of generating the significant adaptive binning descriptor. The specific calculation method is defined as follows:

$$G(\delta) = \exp\left(-\frac{\delta^2}{2\sigma^2}\right) \tag{12}$$

$$\begin{cases} \mathrm{HG}(i, j, o) = \sum\limits_{(x,y)\in B(i,j)} G(D) \cdot \mathrm{SR}_o(x, y) \\ D = \sqrt{(x - x_p)^2 + (y - y_p)^2}. \end{cases} \tag{13}$$

$G(\delta)$ is a 1-D Gaussian function with a standard deviation of σ. $\mathrm{HG}(i, j, o)$ denotes the weighted statistical value corresponding to orientation o in the local histogram located at indexes i, j, this value is computed by weighting the SR of that orientation in the descriptor; i denotes the level of the ring, j refers to the index of subregions within each ring; $B(i, j)$ represents a subregion of the descriptor; D represents the distance of the pixel from the center of the descriptor; (x_p, y_p) is the coordinates of the descriptor's central point.

3) Rotation-Invariant Feature Descriptor: In Section III-C2, we successfully obtained descriptors that can stably describe feature points using SR. However, these descriptors can only match images without rotational differences. Since rotation between images is often unavoidable, we need to estimate the primary orientation of the described region and accordingly rotate the descriptor to this primary orientation to endow the descriptor with rotational invariance. Since the feature point detection and the generation of SR are both obtained through multiorientation log-Gabor filtering, we can use its inherent directional characteristics to determine the primary orientation. Fig. 7 illustrates the procedure of generating feature description vectors with rotational invariance.

First, Fig. 7(a) illustrates the process where the weighted SR is calculated for all pixels within the descriptor range across various orientations to identify the peak index. Then, based on the peak index, we further determine the peak orientation. The method for calculating the peak orientation is as follows:

$$\begin{cases} po = \dfrac{\pi}{O}(o_{\max} - 1) \\ o_{\max} = \underset{o\in[1,O]}{\mathrm{argmax}}(va_o) \\ va_o = \sum\limits_{(x,y)\in \mathrm{CR}} G(D) \cdot \mathrm{SR}_o(x, y). \end{cases} \tag{14}$$

po represents the peak orientation; $o_{\max}$ represents the index of the peak orientation; argmax denotes the indexing operation that takes the maximum value; va_o represents the total of the weighted values of SR_o for each pixel in the descriptor range; CR is the range of the descriptor.

In fact, the peak orientation can directly serve as the primary orientation. However, since our constructed log-Gabor filter only covers the range $[0, \pi)$, the primary orientation obtained from the peak orientation is also limited to this range. To extend to the full range of $[0, 2\pi]$, as depicted in Fig. 7(b), a dominant region can be established as a reference to solidify the relative relationship with the primary orientation. The specific method involves dividing the descriptor region into four quadrants and calculating the weighted SR for the peak orientation filtering in each quadrant. The quadrant with the maximum value is determined as the dominant region. This dominant region is then used to determine whether π should be added to the peak orientation po, as illustrated in Fig. 7(c). Since the peak orientation only occurs in quadrants A_1 and B_1, it can be deduced that there are two scenarios: both the peak orientation and the dominant region are in quadrants on the same diagonal, and the peak orientation and dominant region are in quadrants not on the same diagonal. For cases where both the peak orientation and the dominant region are in quadrants on the same diagonal, we ensure that the primary orientation and the dominant region are in the same quadrant.

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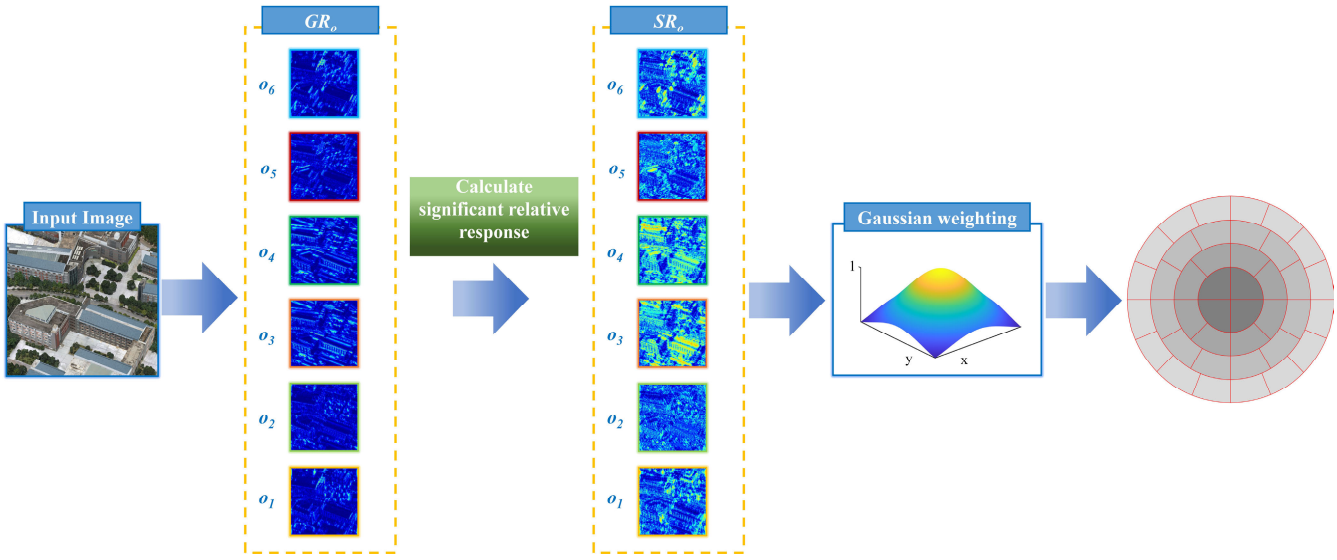

Fig. 6. Development of the significant adaptive binning.

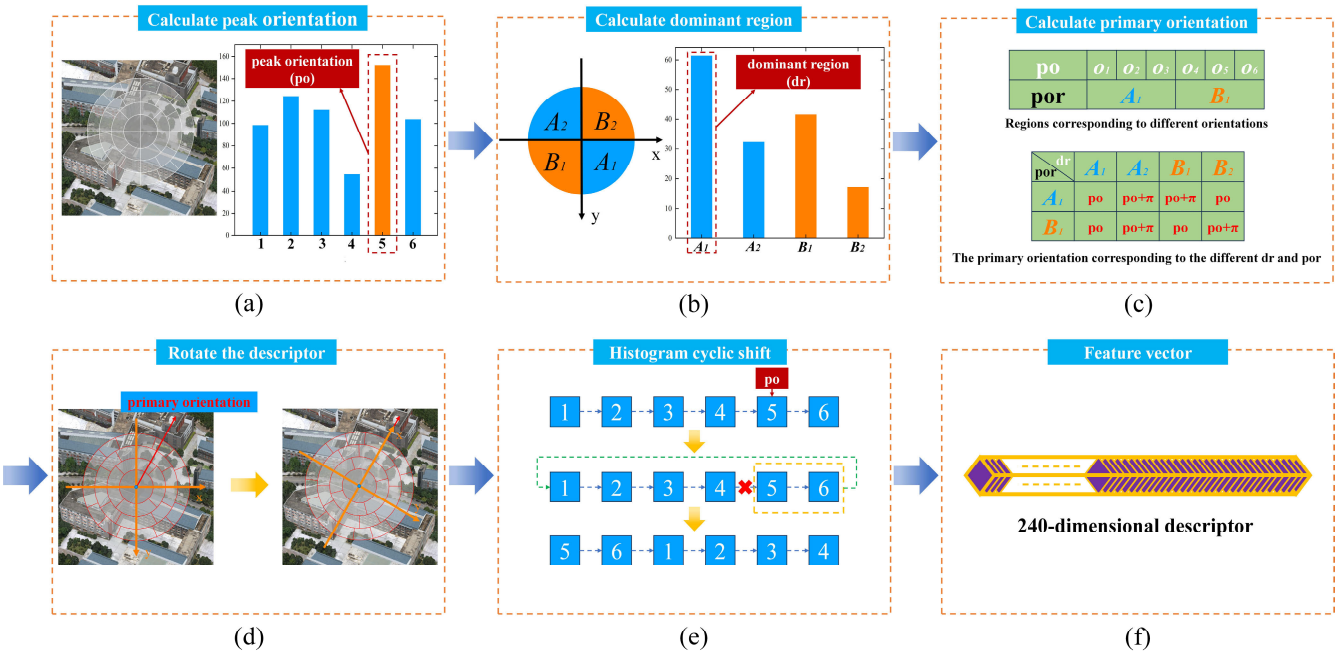

Fig. 7. Generate description vectors with rotation-invariant. (a) Build a histogram to get the peak orientation. (b) Divide the description area into four quadrants and calculate the dominant region. (c) Criteria for calculating primary orientation. (d) Rotate the descriptor to the primary orientation. (e) Adjust the position of the different orientations in the histogram. (f) Final feature vector.

For cases where the peak orientation and the dominant region are not on the same diagonal, we ensure that the primary orientation is in the clockwise direction of the dominant region. This procedure provides us with a method to calculate the primary orientation within the $[0, 2\pi]$ range. To improve the descriptor's robustness, we also consider orientations with values greater than 80% of the peak orientation as auxiliary peak orientation.

As shown in Fig. 7(d), after obtaining the principal orientation, the x-axis of the descriptor coordinate system can be rotated to align with the primary orientation, allowing for the construction of a rotationally invariant descriptor using the method outlined in the previous section. Because the descriptor has been rotated, before calculating the descriptor vector, we need to readjust the positions of the SR from different orientations in the histogram based on the peak orientation, as depicted in Fig. 7(e). The detailed computation method is given as follows:

$$o_{\text{new}} = \begin{cases} o - o_{\max} + 1, & o \geq o_{\max} \\ o - o_{\max} + O + 1, & o < o_{\max}. \end{cases} \tag{15}$$

o_{new} represents the new index order.

Finally, we concatenate each subregion of the descriptor in sequence and apply $L2$ normalization to produce a 240-D feature vector, as depicted in Fig. 7(f).

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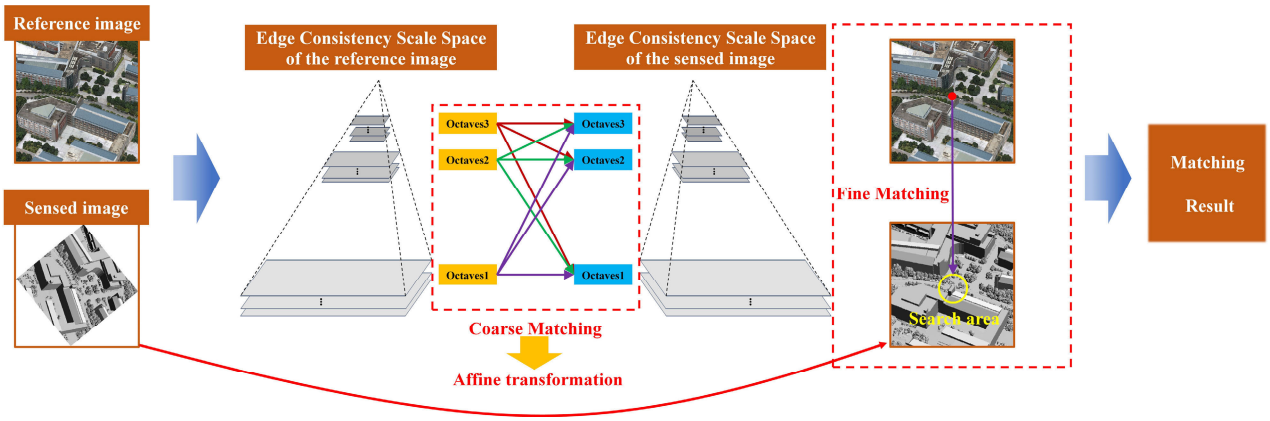

Fig. 8. Coarse to fine matching process.

D. Feature Matching

Fig. 8 shows the matching process from coarse to fine. To maximize the success rate of matching images with scale differences, we adopt a strategy of matching feature vectors of images of different sizes separately rather than matching all descriptors together. Specifically, we perform independent brute force matching for each octave in the image pyramid and use the FSC algorithm to eliminate mismatches. Matching results with more than 20 matching points are combined, and the FSC algorithm is applied again to eliminate mismatches. If the number of matching points is less than 20 in each octave, only the matching result with the highest number of matching points is selected. A key advantage of this approach is its ability to efficiently prevent interference between descriptors of images with varying sizes.

Although our algorithm provides solutions for handling scale and rotational differences, it cannot guarantee that the descriptor of each feature point is accurately calculated. Therefore, the number and precision of matched points may be limited. To improve matching precision and boost the quantity of correspondences, we employed a rematching strategy based on affine transformations. First, the affine transformation model between the two images is determined by utilizing the previously matched points as a reference. Then, with this model, an affine transformation is applied to the image to be matched, generating a new version of the image that reduces the geometric differences between the two images. Simultaneously, feature points detected in each layer of the first octave on the original image to be matched are mapped onto the new image to be matched. Subsequently, we recalculate the feature descriptors for each feature point on both the reference image and the new image to be matched. Notably, since the images have been preprocessed through affine transformations, this calculation does not require estimating the primary orientation. The positional differences between each pair of corresponding points should not be significant. Therefore, we limit the matching search range to within three times the nonmaximum suppression range. Such a limitation ensures a large number of high-precision matched pairs. After matching is complete, the matched point relationships from the newly transformed image are then mapped back to the original image, and the FSC algorithm is applied to eliminate mismatches further, ensuring the precision of the final results.

IV. Experiments and Analysis

In this section, to assess the effectiveness of our proposed ECSS matching method, we prepared five different types of MRSI datasets for both qualitative and quantitative evaluation. Additionally, we evaluated the ECSS method alongside nine advanced matching methods, including two deep learning-based methods, ReDFeat and MINIMA-LG, and seven handcrafted-based methods, RIFT, MS-HLMO, SRIF, WSSF, POS-GIFT, OFM, and GLS-MIFT. To validate the algorithm's applicability in practical remote sensing tasks, two pairs of real-scenario optical-SAR images were employed for performance evaluation. Comparative experiments with the advanced area-based methods SFOC were conducted to rigorously verify the effectiveness of the proposed method in engineering applications. All methods used for comparison were implemented using either the original code from their respective authors or publicly available software resources. All experiments were performed on an MSI Titan 16HX laptop featuring an Intel i9-13980HX CPU (2.20 GHz), 64 GB of RAM, and Windows 11 × 64 OS.

A. Datasets and Experimental Settings

We selected five different types of data for the experiment, including optical-optical, optical-map, optical-depth, optical-SAR, and optical-infrared. Each type consists of ten pairs of images, totaling 50 pairs for the test dataset. To increase the experiment's challenge, we introduced scale variations from 0.5 to 2× and rotational differences from 0° to 360° in each pair of images. Fig. 9 shows several sample image pairs from the datasets. For each pair of images, we manually annotated 10–20 matching points and used these points to compute the affine transformations between the two images. The affine transformation was used as the ground truth to evaluate the accuracy of the automatically matched points' locations.

The number of correct matches (NCMs), the root mean square error (RMSE), and the success rate (SR) are used as quantitative evaluation metrics. NCM is used to measure matching capability, with higher NCM indicating stronger matching ability. The definition of a correct match is provided as (16), where pairs of matching points with errors of less than three pixels are considered correctly matched. RMSE serves as an indicator of matching accuracy, where a lower RMSE value

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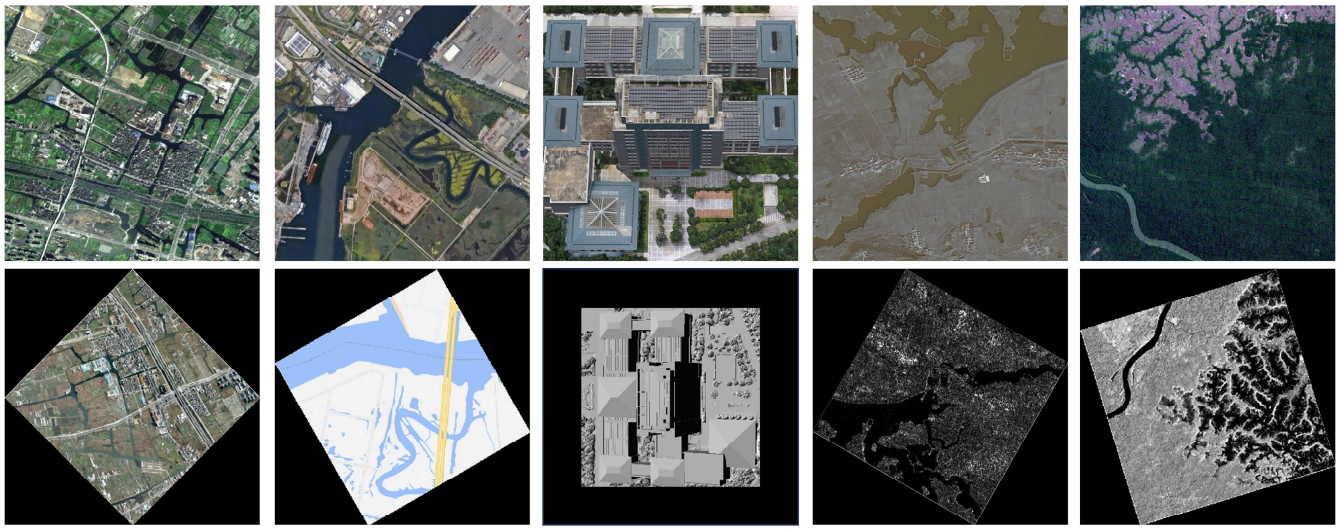

Fig. 9. Samples of MRSI datasets. (a) Optical-optical. (b) Optical-map. (c) Optical-depth. (d) Optical-SAR. (e) Optical-infrared.

corresponds to greater precision in the results; the definition of RMSE is shown as (17). SR evaluates the robustness of the matching process, where a higher SR signifies increased reliability and stability in the matching performance. A match is considered failed when the NCM is less than 10 or the RMSE is larger than five pixels

$$\text{correct match} = |\text{ref}(x, y) - T_{\text{truth}} * \text{sen}(x, y)| < 3. \quad (16)$$

$\text{ref}(x, y)$ denotes the point on the reference image; T_{truth} represents the transformation matrix computed from the ground truth; $\text{sen}(x, y)$ denotes the matching feature point of $\text{ref}(x, y)$ on the sensed image

$$\text{RMSE} = \sqrt{\frac{1}{\text{NOM}} \sum_{i=1}^{\text{NOM}} \left[\left(x_i - x_i'\right)^2 + \left(y_i - y_i'\right)^2 \right]}. \quad (17)$$

NOM refers to the number of output matches; (x_i, y_i) denotes the position on the reference image; (x_i', y_i') represents the position of the matching point corresponding to (x_i, y_i) on the sensed image after the correct matrix transformation.

B. Parameter Setting Tests

In the ECSS algorithm, four key parameters need to be determined through testing to achieve optimal matching results. These parameters include: the number of layers L in each octave of the pyramid, the initial standard deviation σ_0, the descriptor radius R, and the distance d for nonmaximum suppression. Other parameters in the algorithm are set based on existing research.

L denotes the number of layers in each octave of the pyramid. Theoretically, the more layers L there are the greater the continuity of scales, enabling the extraction of a higher number of feature points within an octave. However, an excessive number of layers will add to the algorithm's computational load. The initial standard deviation σ_0 determines the smoothness of each layer of the image; too high σ_0 value may result in excessive smoothing, thereby losing effective feature information. R represents the radius of the descriptor; the greater the value, the more comprehensive and detailed the information encapsulated within the descriptor, thereby increasing the distinctiveness between descriptors. However, too small an R value may lead to a lack of uniqueness in descriptors, and too large a value might cause matching failures due to excessive differences in descriptors of homonymous points. d is the distance used for nonmaximum suppression; a larger d value means that the retained feature points have stronger responses. However, too large a d value may result in too few remaining feature points, and too small a d value might cause feature points to cluster too closely, increasing matching errors.

To determine the optimal parameter settings, we adjusted specific parameters during testing while keeping others fixed. For performance evaluation, we used NCM, RMSE, and SR as the metrics. Table I displays the detailed results of the experiments. Specifically, we adjusted the value of L from 1 to 4, set the range of the initial standard deviation σ_0 from 1.0 to 1.4, increased the descriptor radius R from 32 to 56, and adjusted the nonmaximum suppression distance d from 4 to 10. We found that the NCM value peaked when L was set to 4; however, the RMSE also increased. Given the marginal increase in NCM compared to $L = 3$, and to control computational costs, we ultimately decided to set L to 3. When the initial standard deviation σ_0 was set at 1.2, all evaluation metrics were optimal, so we chose $\sigma_0 = 1.2$. For the descriptor radius R, when set at 48, SR reached 100%, and RMSE was also at its best, so we set R to 48. Although increasing the nonmaximum suppression distance d decreased the NCM value, RMSE also decreased. Taking into account both NCM and RMSE, d is set to 8.

C. Results and Analysis

To evaluate the performance of the ECSS method we developed, we conducted a comparison with seven cutting-edge multimodal matching methods, including ReDFeat, MINIMA-LG, RIFT, MS-HLMO, SRIF, WSSF, POS-GIFT, OFM, and

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TABLE I
DETAILED EXPERIMENTAL RESULTS OF ECSS UNDER DIFFERENT PARAMETER SETTINGS

Tested parameter	Parameter setting				NCM↑	RMSE↓	SR(%)↑
	L	σ_0	R	d			
L	1	1.2	48	8	1533.62	2.36	88.00
	2	1.2	48	8	1726.85	2.46	**100.00**
	3	1.2	48	8	1788.24	**2.09**	**100.00**
	4	1.2	48	8	**1796.51**	2.17	**100.00**
σ_0	3	1.0	48	8	1738.11	2.37	96.00
	3	**1.2**	48	8	**1788.24**	**2.09**	**100.00**
	3	1.4	48	8	1698.56	2.16	**100.00**
	3	1.6	48	8	1572.53	2.38	96.00
R	3	1.2	32	8	1692.49	2.79	92.00
	3	1.2	40	8	**1831.73**	2.21	96.00
	3	1.2	**48**	8	1788.24	**2.09**	**100.00**
	3	1.2	56	8	1736.57	2.13	**100.00**
d	3	1.2	48	4	**1815.34**	2.56	**100.00**
	3	1.2	48	6	1794.95	2.33	**100.00**
	3	1.2	48	**8**	1788.24	**2.09**	**100.00**
	3	1.2	48	10	1768.65	2.17	**100.00**

GLS-MIFT. Table II presents the average results of these methods across a dataset of 50 images for various metrics.

Table III shows the average results for each metric on different types of images. When calculating the average RMSE, we uniformly accounted for failed matches as five pixels. To visually demonstrate the test results of each method, Fig. 10(a) illustrates the RMSE for each pair of images, with the symbol "$+\infty$" indicating an RMSE exceeding five pixels. Fig. 10(b) displays the average NCM for each method across different types of images.

While ReDFeat and MINIMA-LG, as deep learning-based methods, possess multimodal image matching capabilities, their performance was significantly constrained in our experimental dataset. This limitation stems from their algorithm architectures lacking essential rotation-invariant features, as they were not adequately designed or trained to account for substantial rotational variations between images. Consequently, their SR were merely 8% and 10%, with NCM of 4.56 and 41.46 and RMSE reaching 4.90 and 4.73 pixels, respectively.

RIFT addresses the NRD in multimodal images by calculating the MIM. This method stably describes the structural features of images, because even though the response values of log-Gabor filters may be unstable across different images, the orientation of their response peaks usually remains consistent. However, among all handcrafted-based methods compared, RIFT shows the poorest performance. Across all types of data, its SR does not exceed 20%, with an average NCM of 31.26 and an RMSE as high as 4.78 pixels. This is mainly because each pair of images in our test data exhibits scale differences, and RIFT is not designed to be scale-invariant.

MS-HLMO outperforms RIFT by a small margin in terms of performance, with an average SR of 34%, an average NCM of 17.92, and an average RMSE of 4.64 pixels. These figures indicate that MS-HLMO fails to effectively match the majority of images. The algorithm utilizes GGLOHs to construct feature descriptors, aiming to counteract the effects of NRD. However, since the method continues to use the HARRIS algorithm for feature point detection in the original images, this detection approach is significantly affected by NRD, resulting in a low repeatability rate of feature points detected between the two images, thereby leading to suboptimal test results.

SRIF generates an intermediate state image for feature detection and description by computing the local intensity binary transform (LIBT). This approach aims to enhance the similarity between multimodal images. However, the computation of LIBT is based solely on pixel intensity, and the image pyramid is constructed without applying any filtering throughout the process, making the algorithm susceptible to noise interference. Consequently, in data types with poor image quality and high noise, such as Optical-SAR and Optical-Infrared, the SR is below 50%. Overall, SRIF has an average SR of 58%, an average NCM of 53.58, and an average RMSE of 3.93 pixels.

WSSF utilizes structured random forests to identify the edge structures of images and based on this, constructs pointwise shape-adaptive texture filters (PSATF) to establish an image pyramid. At each level of the pyramid, the method computes a structural significance map (SSM), which effectively preserves the structural features of the image and significantly reduces the impact of NRD. However, this approach heavily relies on the edge detection performance of structured random forests, in areas with blurred edges or complex structures, the loss of detail features can lead to poor matching results. WSSF has an average SR of 66%, an average NCM of 30.86, and an average RMSE of 3.88 pixels. WSSF has a lower NCM because it calculates descriptors on the SSM, which is relatively sparse, resulting in descriptors with reduced discriminative ability.

POS-GIFT achieves an SR of over 70% in all data types except for Optical-Optical, where the SR is only 30%. This significant discrepancy is mainly due to POS-GIFT's poor rotational invariance at certain specific angles, which leads to a sharp decline in matching capability. Due to the optimization of matching results through the POS strategy, the average NCM reaches 684.24, with an average SR of 74% and an average RMSE of 3.17 pixels. These results indicate that POS-GIFT possesses strong matching capabilities for the majority of images.

Similar to RIFT, OFM also uses the maximal index to construct descriptors to counteract NRD in multimodal images. OFM introduces a Gaussian pyramid to achieve scale invariance and employs a more efficient computation method to

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TABLE II
QUANTITATIVE COMPARISON RESULTS FOR ALL DATASETS

Metric	ReDFeat	MINIMA-LG	RIFT	MS-HLMO	SRIF	WSSF	POS-GIFT	OFM	GLS-MIFT	ECSS
NCM	4.56	41.46	31.26	17.92	53.58	30.86	684.24	598.42	1122.92	**1788.24**
RMSE	4.90	4.73	4.78	4.64	3.93	3.88	3.17	2.70	2.53	**2.09**
SR(%)	8.00	10.00	14.00	34.00	58.00	66.00	74.00	92.00	94.00	**100.00**

TABLE III
QUANTITATIVE COMPARISON RESULTS ON DIFFERENT DATASETS

Datasets	Metric	ReDFeat	MINIMA-LG	RIFT	MS-HLMO	SRIF	WSSF	POS-GIFT	OFM	GLS-MIFT	ECSS
1. Optical-Optical	NCM	4.70	0.00	14.40	2.40	38.60	12.20	250.50	549.50	1119.40	**1944.10**
	RMSE	4.87	5.00	4.98	4.99	4.29	4.55	4.47	2.81	2.45	**2.18**
	SR(%)	10.00	0.00	10.00	10.00	50.00	30.00	30.00	90.00	**100.00**	**100.00**
2. Optical-Map	NCM	0.00	53.90	16.40	30.60	54.20	45.00	812.20	398.60	1057.80	**1720.20**
	RMSE	5.00	4.67	4.85	4.30	3.67	3.57	2.72	2.75	2.84	**2.16**
	SR(%)	0.00	10.00	10.00	70.00	80.00	90.00	90.00	100.00	90.00	**100.00**
3. Optical-Depth	NCM	8.50	33.90	43.50	5.60	96.70	35.90	771.10	512.90	924.80	**1568.30**
	RMSE	4.78	4.79	4.86	4.95	3.07	3.81	2.78	2.53	2.41	**1.99**
	SR(%)	10.00	10.00	20.00	30.00	90.00	70.00	**100.00**	**100.00**	**100.00**	**100.00**
4. Optical-SAR	NCM	6.40	26.70	75.10	39.20	39.50	24.50	651.40	633.20	1057.60	**1659.10**
	RMSE	4.88	4.81	4.74	4.42	4.22	4.18	2.97	2.86	2.76	**2.25**
	SR(%)	10.00	10.00	10.00	40.00	40.00	60.00	80.00	80.00	80.00	**100.00**
5. Optical-Infrared	NCM	3.20	92.80	6.90	11.80	38.90	36.70	936.00	897.90	1455.00	**2049.50**
	RMSE	4.97	4.40	4.78	4.54	4.41	3.29	2.92	2.54	2.21	**1.85**
	SR(%)	10.00	20.00	20.00	20.00	30.00	80.00	70.00	90.00	**100.00**	**100.00**

ensure rotational invariance. Across the entire dataset, OFM achieves an average SR of 92%, indicating successful matching of most images. Additionally, its average NCM is 598.42, with an average RMSE of 2.70 pixels. Nonetheless, the use of maximal index to build descriptors may significantly reduce discriminability in images with weak textures or highly similar regions, potentially leading to matching failures.

GLS-MIFT also employs the method of using the maximal index to construct descriptors. This method has demonstrated outstanding performance across all test datasets, with an average SR of 94%, an average NCM of 1122.92, and an average RMSE of 2.53 pixels, ranking second only to ECSS among all methods compared. However, similar to OFM, GLS-MIFT also fails to match some images with highly similar textures.

ECSS achieved successful matches on all 50 test image pairs, with an average SR of 100%, an average NCM of 1788.24, and an average RMSE of only 2.09. These metrics are the best among the eight methods. These results indicate that ECSS not only effectively resists NRD in multimodal images but also handles variations in scale and rotation well. Consequently, ECSS can achieve high-precision matching of multimodal images.

Figs. 11–20 show the matching outcomes of the ten methods applied to sample images. As shown in Figs. 11 and 12, both ReDFeat and MINIMA-LG failed to produce successful matches on the exemplar datasets due to the lack of rotation invariance. Fig. 13 shows RIFT's matching outcomes, due to its lack of scale invariance, RIFT failed to successfully match most images. Fig. 14 presents the matching outcomes of MS-HLMO, which uses HARRIS for feature point detection, resulting in a lower repeatability rate of feature points on the images and, thus, fewer matching points. Fig. 15 illustrates that although SRIF mitigates the impact of NRD between images via LIBT, its matching results lack robustness on images with poor imaging effects and indistinct features. Fig. 16 shows the matching outcomes of WSSF, which constructs scale space using PSATF, effectively highlighting the structural features

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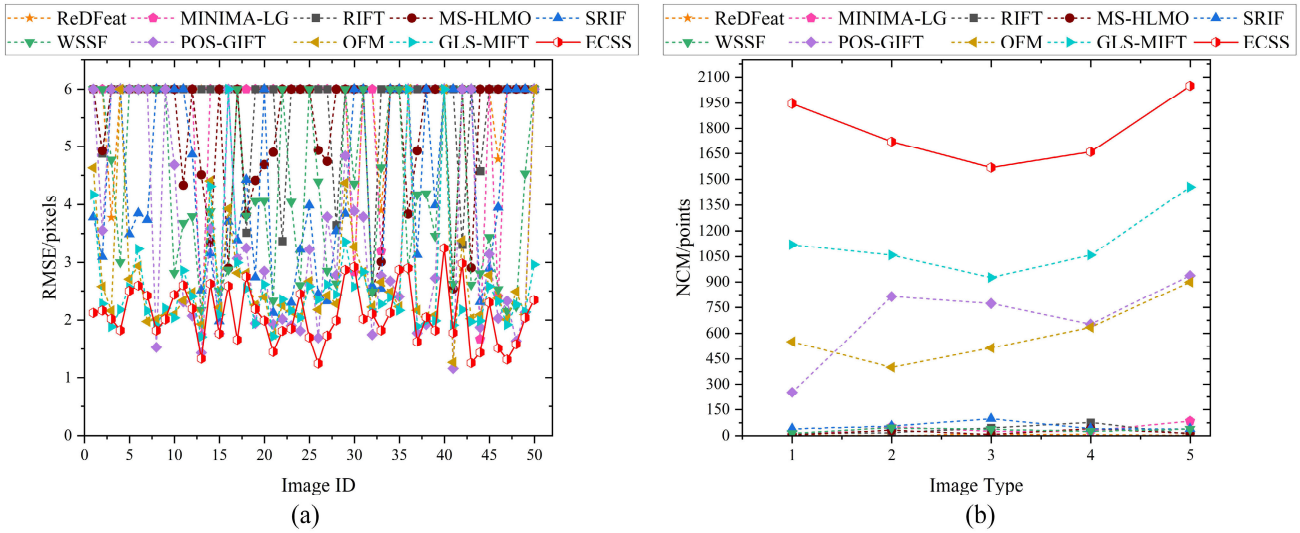

Fig. 10. Test results of the ten methods. (a) RMSE of the ten methods on each pair of images. (b) Average NCM of the ten methods on each type of dataset.

of images. This enables successful matching on images with clear edge information, but performance may significantly decline or even fail on images with complex structures or less distinct features. Fig. 17 reveals that POS-GIFT's matching outcomes obtain numerous matching points on most images, yet its robustness against rotational effects is lacking, potentially leading to matching failures at certain angles. Figs. 18 and 19, respectively, show the outcomes of OFM and GLS-MIFT, which construct descriptors using an indexed image approach, exhibiting high robustness and achieving very good matching results, although they may fail on images with homogeneous content. Fig. 20 illustrates the ECSS method's successful alignment of multimodal images with substantial geometric variations, highlighting its strong capabilities in remote sensing image matching.

Fig. 21 displays the matching outcomes of the ECSS method on the remaining 45 images. Notably, despite facing challenges with significant geometric and radiometric differences, ECSS successfully identified numerous correspondences on these images. This not only demonstrates its excellent matching capabilities but also reflects ECSS's outstanding performance in handling complex multimodal images, effectively overcoming numerous challenges in the multimodal remote sensing image matching field.

D. Average Running Time

Table IV presents the average runtime of each algorithm on the dataset. ReDFeat and MINIMA-LG, deep learning-based methods, utilize CUDA for acceleration, resulting in shorter runtimes. SRIF minimizes runtime among all handcrafted-based methods by using statistical pixel intensity values to eliminate NRD without filtering operations. Among all algorithms, MS-HLMO exhibited the longest computation time of 58.74 s. RIFT, WSSF, POS-GIFT, OFM, GLS-MIFT, and ECSS all adopted multiscale and multiorientation filtering frameworks. Although RIFT does not compute scale space, its use of convolution sequence rings for rotational invariance results in the highest time complexity and the longest runtime. WSSF, POS-GIFT, and ECSS demonstrated comparable time costs, with ECSS showing marginally increased runtime due to its pyramid-level matching strategy. OFM and GLS-MIFT exhibited relatively reduced efficiency due to their rotational augmentation strategies for enhanced rotation invariance. Although multiscale and multidirectional filtering methods yield better matching results, they have higher time complexity. Therefore, the use of multithreading or GPU acceleration techniques could be considered to reduce time costs.

E. Ablation Study on Filters Used in Building Scale Space Pyramid

To assess the efficacy of the proposed edge consistency filtering (ECF), we conducted experiments on all test data using pointwise shape-adaptive texture filtering (PSATF), COF, and GF as alternatives to ECF for constructing scale space, while keeping all other aspects unchanged. Table V shows the experimental results using different filtering methods to build the scale space.

According to the experimental results in Table V, using ECF to construct scale space demonstrates the best performance across all evaluation metrics. ECF determines whether pixels are on the same side of an edge by analyzing the positive or negative responses of the even components of log-Gabor filters. This method effectively removes unnecessary details and noise from the images while avoiding edge blurring, thus maintaining a clear image structure.

PSATF also achieved a 100% SR in our test data. This is because PSATF uses structured random forests to compute image edges, further calculates edge confidence values based on this, and obtains guidance maps for joint bilateral filtering through these edge confidence values. This approach effectively preserves and enhances the edge structures of images during the filtering process. However, compared to ECF, PSATF may blur more details, resulting in a slightly lower

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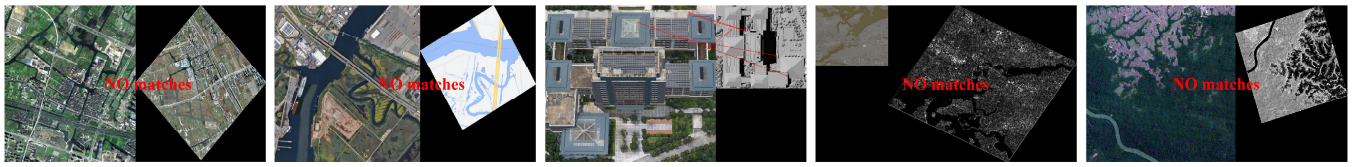

Fig. 11. Matching outcomes of ReDFeat on sample images.

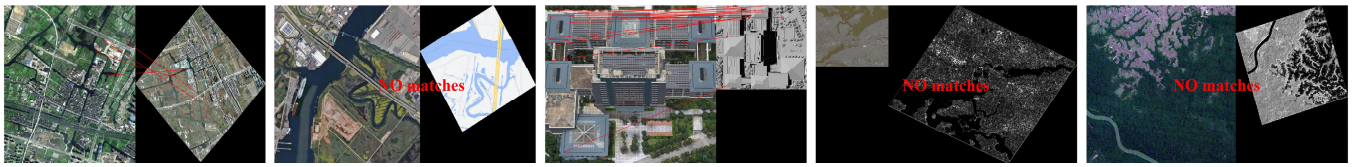

Fig. 12. Matching outcomes of MINIMA-LG on sample images.

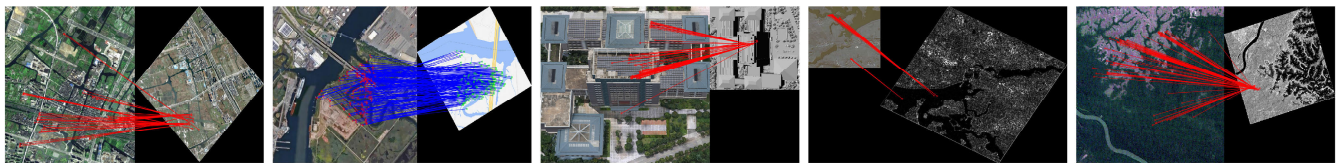

Fig. 13. Matching outcomes of RIFT on sample images.

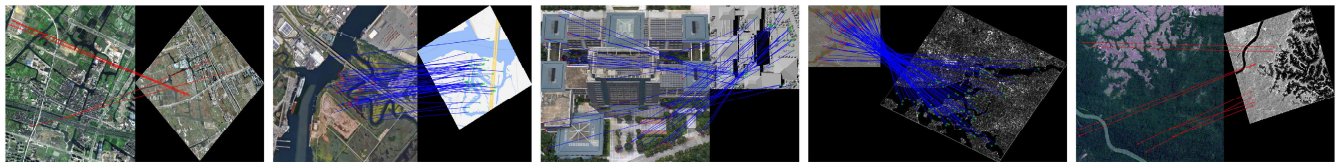

Fig. 14. Matching outcomes of MS-HLMO on sample images.

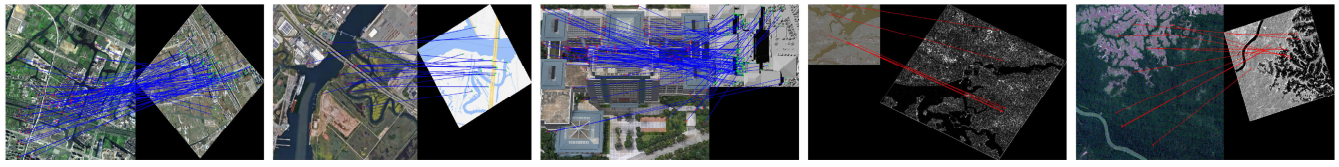

Fig. 15. Matching outcomes of SRIF on sample images.

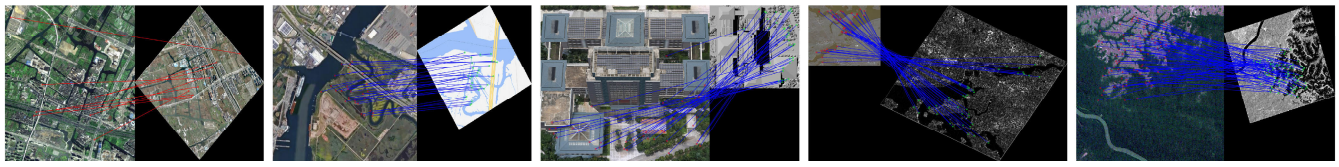

Fig. 16. Matching outcomes of WSSF on sample images.

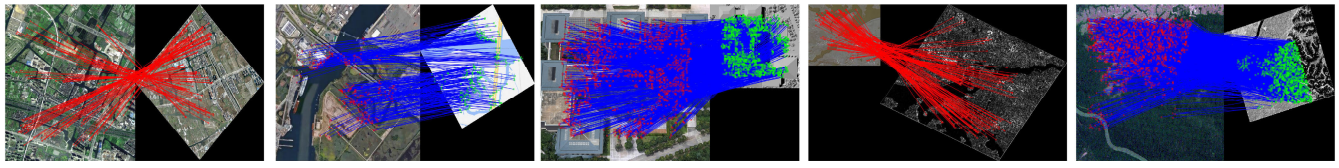

Fig. 17. Matching outcomes of POS-GIFT on sample images.

NCM than the results obtained using ECF to construct scale space. While PSATF requires a separate calculation of the edge confidence map to serve as a guide image, ECF can directly generate the guide image while performing PC calculations, thereby significantly reducing the overall computational load.

COF achieved an SR of 98%, with an RMSE increasing to 2.34, overall performing less effectively than ECF and PSATF.

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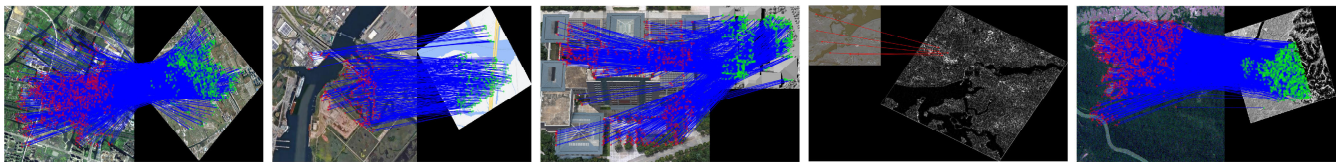

Fig. 18. Matching outcomes of OFM on sample images.

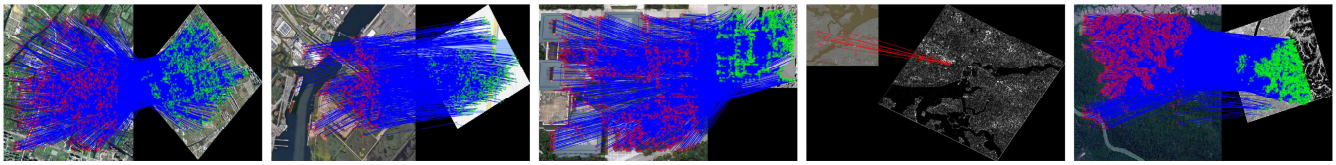

Fig. 19. Matching outcomes of GLS-MIFT on sample images.

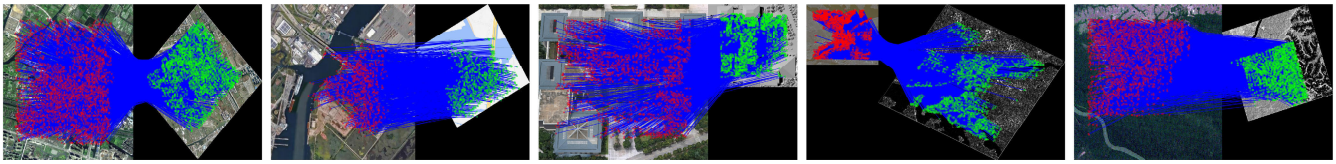

Fig. 20. Matching outcomes of ECSS on sample images.

TABLE IV
AVERAGE RUNNING TIME RESULTS ON ALL DATASETS

Metric	ReDFeat	MINIMA-LG	RIFT	MS-HLMO	SRIF	WSSF	POS-GIFT	OFM	GLS-MIFT	ECSS
Time(s)	0.24	0.13	34.62	58.74	5.56	17.92	17.28	22.58	28.14	19.36

TABLE V
EXPERIMENTAL RESULTS FOR DIFFERENT FILTERS TO BUILD UP THE SCALE SPACE

Metric	ECF	PSATF	COF	GF
NCM	**1788.24**	1562.86	1474.33	1462.58
RMSE	**2.09**	2.26	2.34	2.86
SR(%)	**100.00**	**100.00**	98.00	96.00

This is primarily because, unlike ECF and PSATF, which generate guide images through the structural information of images, COF only generates guide images by calculating the probability of co-occurrence between pixels. This approach results in COF often overlooking numerous fine edges and struggling to effectively preserve edges in images with complex structures.

In testing, GF performed the worst across all evaluation metrics, achieving only a 96% SR and an RMSE of 2.86 pixels. This result was anticipated, as GF does not incorporate any edge-preserving measures. During the process of constructing scale space, the clarity of edges is compromised, which negatively affects the accuracy of feature point localization.

To demonstrate the scale invariance of the ECSS method, Fig. 22 illustrates the matching outcomes for a pair of optical-map images with scale variations ranging from 0.4 to 2.6×. Fig. 23 shows the NCM results under different scale differences, where ECSS-COARSE represents the result of coarse matching only. As the scale difference increases, the NCM decreases, but ECSS-COARSE still yields a sufficient number of matching points to meet the registration requirements. The addition of the fine matching step significantly increased the NCM count for ECSS, with the NCM remaining stable between 2450 and 2700, except for the case with a 0.4× scale difference. At a 0.4× scale difference, the smaller size of the images to be matched results in fewer detected feature points, leading to a reduction in the number of matching points, although it remains above 2000. It can be seen that ECSS demonstrates good scale invariance and validates the importance of the fine matching step.

F. Ablation Study on Significance Relative Response

To validate the robustness of our proposed significant relative response (SR) in constructing feature descriptors, we conducted a comparative experiment where the rest of the algorithm remained unchanged, but the basis for constructing feature descriptors was switched from SR to log-GRs and MIM, resulting in the variants ECSS-GR and ECSS-MIM, respectively. Fig. 24 displays the test results using these methods for constructing feature descriptors.

SR effectively maximizes the information from log-Gabor filtering in each orientation, while avoiding interference

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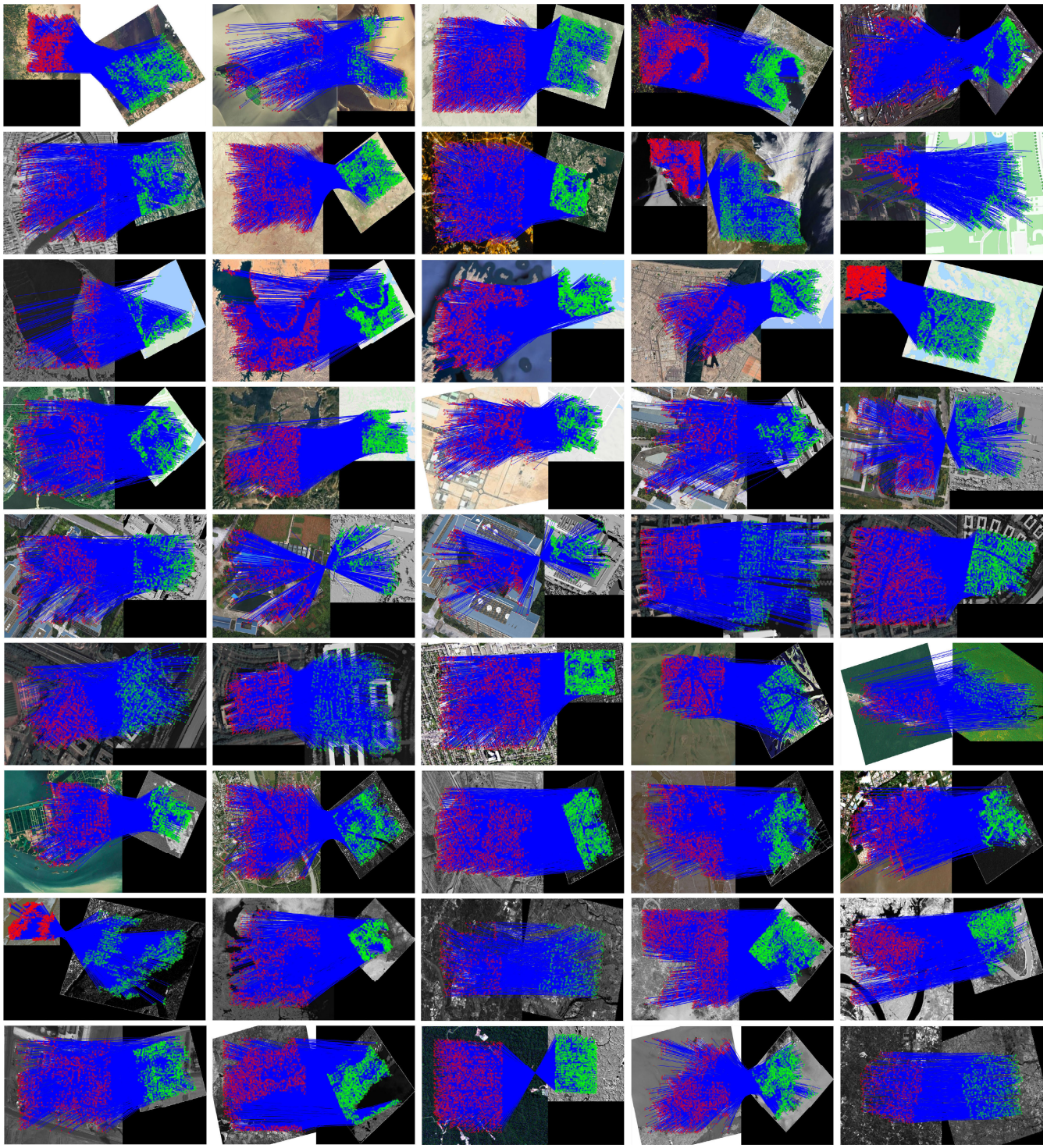

Fig. 21. Matching outcomes of the ECSS method.

from local extrema in the descriptors and emphasizing the relative importance of responses in different orientations. This approach significantly improves the robustness of the matching process. Consequently, as illustrated in Fig. 24, the method utilizing SR (ECSS) is the only one that successfully matched all test images.

The success rate for matches using descriptors constructed with GR is the lowest due to the instability of GR values across different images, making the descriptors generated unable to accurately describe corresponding points, thus reducing the success rate of matches. However, we found that in the image pairs that were successfully matched, descriptors constructed with GR often achieved lower RMSE. This may be because using GR directly to construct descriptors, although it reduces the similarity of descriptors for corresponding points, simultaneously maximizes the differences between descriptors of different feature points, thereby increasing the precision of matching.

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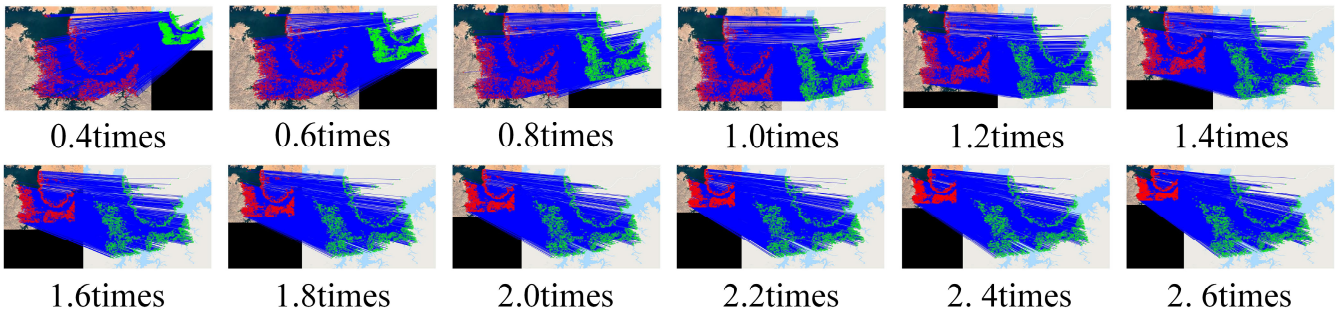

Fig. 22. Matching outcomes under different scale differences of the ECSS method.

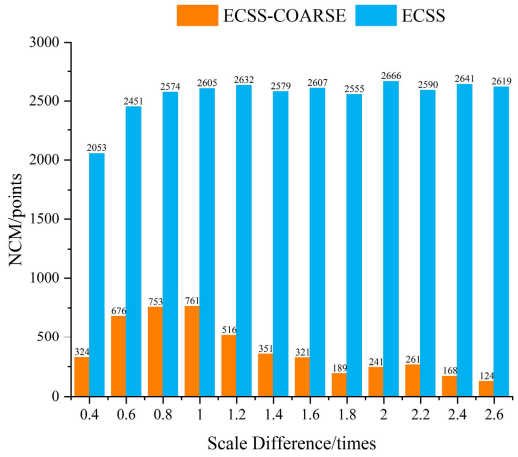

Fig. 23. Matching results for different scale differences.

MIM constructs descriptors by using the directional index of the maximal response in log-Gabor filtering, effectively mitigating the effects of NRD in images. Compared to constructing descriptors directly from filter values, MIM offers greater stability. However, this method does not utilize all the multiorientation information from log-Gabor filtering, and the discriminative power of the descriptors may be too low on images with similar textures or homogeneous features, resulting in overall performance that is inferior to descriptors constructed using SR.

G. Ablation Study on Rotation-Invariant

To validate our proposed rotation invariance calculation method, we selected a pair of images and generated 36 images at 10° intervals from 0° to 350° to test matching performance at different angles. The tests were primarily evaluated through the NCM of matches. The original algorithm refined the matching results using a re-matching strategy with affine transformations, leading to almost no significant differences in the final match outcomes across different angles. Therefore, to solely evaluate the algorithm's matching performance in rotational scenarios, we removed the fine matching strategy and only discussed ECSS-COARSE. Additionally, to further explore the effects of different primary orientation calculation strategies, we set up three strategies for comparison: ECSS-SIFT, ECSS-SURF, and ECSS-MIFT. In all comparative

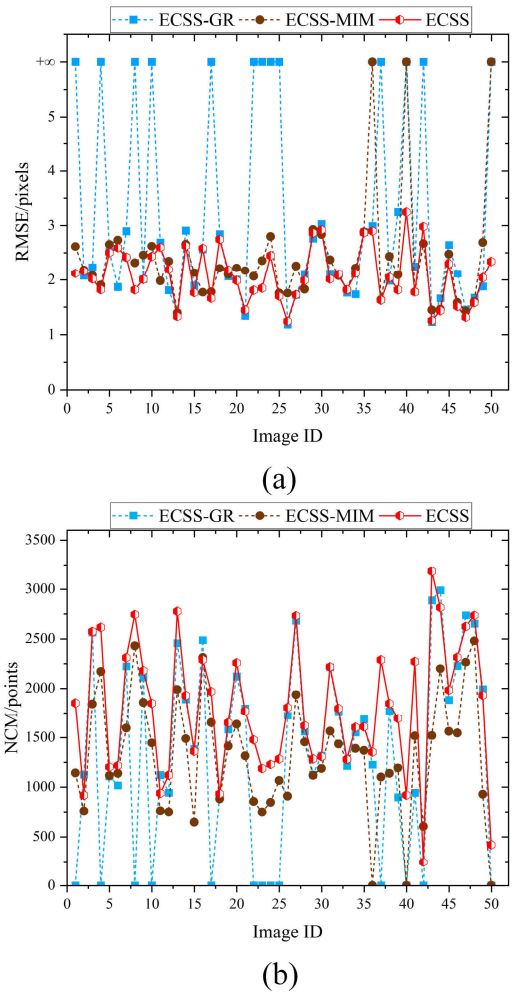

Fig. 24. Test results using different methods for constructing feature descriptors. (a) RMSE of the three methods on each pair of images. (b) NCM of the three methods on each pair of images.

methods, we only modified the calculation of the primary orientation, while all other aspects remained unchanged. Fig. 25 shows the results of calculating the primary orientation using different methods. The primary orientation calculation methods compared are as follows.

1) ECSS-SIFT employs a method similar to SIFT for calculating the primary orientation. To reduce the impact of NRD, calculations are not performed on the original image; instead, gradient histograms around feature

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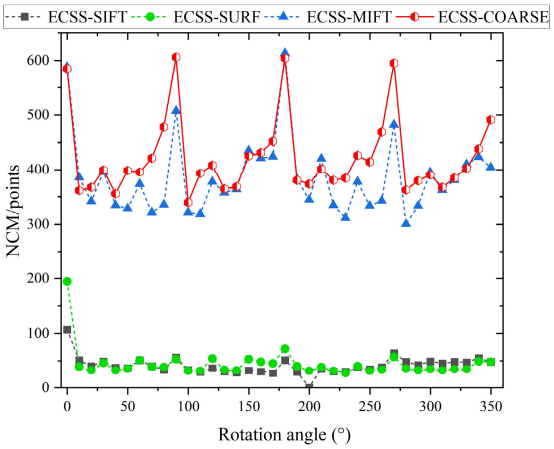

Fig. 25. Matching results for different ways of calculating primary orientation.

TABLE VI
QUANTITATIVE COMPARISON FOR MATCHING RESULTS OF OPTICAL-SAR IMAGE PAIRS

Datasets	Method	RMSE (pixel)
Shigatse City	ECSS	4.76
	SFOC	4.21
Dujiangyan City	ECSS	3.69
	SFOC	failed
Dujiangyan City*	ECSS	2.37
	SFOC	2.42

points are compiled on the maximum moment. Subsequently, the orientation with the highest value in the histogram is chosen as the primary orientation.

2) ECSS-SURF utilizes a method similar to SURF to calculate the primary orientation. Specifically, the region surrounding the feature points is divided into multiple sectors, and the Haar wavelet response within each sector is tallied. The primary orientation is defined by the sector that has the highest response value. Similar to ECSS-SIFT, to minimize the impact of NRD, all these calculations are conducted on the maximum moment.
3) ECSS-MIFT employs a method similar to GLS-MIFT to calculate the primary orientation. Essentially, both GLS-MIFT and ECSS utilize the directional information of multiorientation filters themselves to determine the primary orientation. The key distinction between them is the strategy used to determine whether to add π to the peak orientation. In GLS-MIFT, this decision is based solely on the descriptor's dominant region: if the dominant region is in areas A_1 and B_1, the peak orientation is considered the primary orientation; if it is in areas A_2 and B_2, the primary orientation is the peak orientation plus π.

From Fig. 25, it is evident that despite using SIFT and SURF methods on the maximum moment map to mitigate the impact of NRD, the results are still unsatisfactory. At most rotation angles, the NCM values are below 100. This is primarily because, although the maximum moment map significantly displays the structural features of the images, their sparse content leads to instability in the primary orientation calculated on this basis.

Both ECSS-MIFT and ECSS-COARSE achieved very good results. We noticed that the matching results of ECSS-COARSE exhibit a clear 90° periodicity. This is because rotating the image alters the count of valid pixels, which impacts the identification of feature points and the accuracy of feature descriptions. The impact is minimized when the rotation angle is an exact multiple of 90°. Additionally, since our calculated dominant regions are also divided into four quadrants at 90° intervals, theoretically, when the rotational difference between two images is a multiple of 90°, the matching results should be very similar. However, the NCM of ECSS-MIFT at 90° and 270° are significantly lower than those of ECSS-COARSE. Since the calculated peak orientations are the same, this is clearly due to the strategy of whether to add π. Compared to merely determining whether to add π based on the position of the dominant region, the strategy of using the relative position of the peak orientation to the dominant region to decide whether to add π can reduce the occurrence of erroneously calculated descriptors. Fig. 26 displays the matching outcomes of the complete ECSS algorithm at various rotation angles, applied to a pair of optical-SAR and optical-infrared images.

H. Optical-SAR Image Registration and Discussion With Area-Based Method

To provide further validation of the effectiveness of ECSS in practical engineering tasks such as disaster analysis, two sets of multitemporal and multimodal remote sensing data were selected for registration experiments. Fig. 27 shows the experimental data.

1) *Shigatse City, Tibet Autonomous Region, China:* Pre-earthquake optical image (BJ-2 satellite, 0.8-m resolution); post-earthquake SAR image (PIESAT-2 satellite, 2-m resolution).
2) *Dujiangyan City, Sichuan Province, China:* Pre-earthquake optical image (IKONOS-2 satellite, 1-m resolution); postearthquake SAR image (COSMO-SkyMed satellite, 1-m resolution).

Due to the large width of the images, direct matching is infeasible. Therefore, a block matching strategy is adopted. The image is divided into several blocks based on geographic coordinate information, and independent matching of the overlapping areas of each block is performed. The overall registration is achieved by integrating the matched points from all subregions.

Fig. 28 shows the comparison of the red inspection area in Fig. 27 before and after registration. Before registration, there was a noticeable shift in the image, which was largely eliminated after registration, allowing the edges within the inspection area to align nearly perfectly.

To quantitatively evaluate the registration results on these two datasets, we selected no fewer than 15 prominent feature

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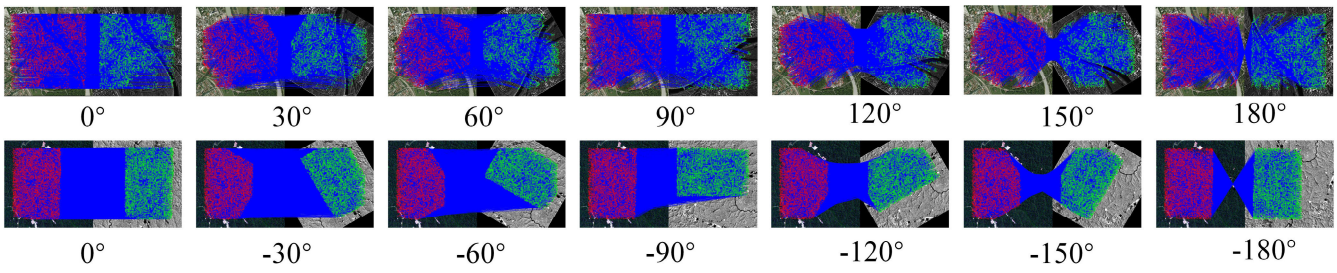

Fig. 26. Matching outcomes under different rotation differences of the ECSS method.

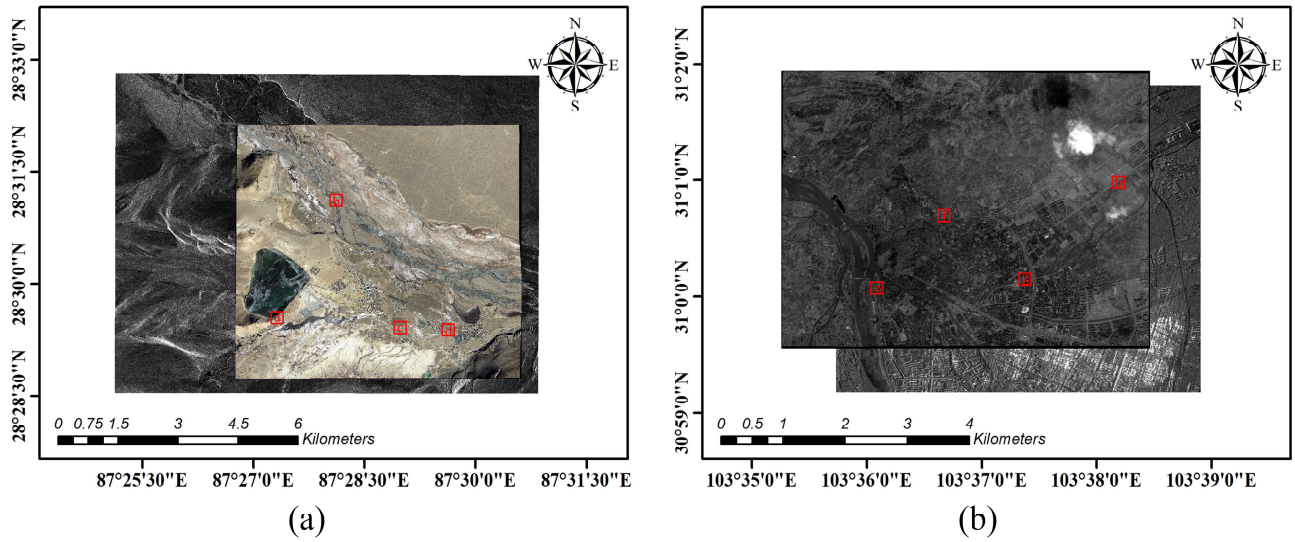

Fig. 27. Optical-SAR data to be registered, with the red rectangle indicating the inspection area. (a) Optical-SAR image pair of the earthquake area in Shigatse City. (b) Optical-SAR image pair of the earthquake area in Dujiangyan City.

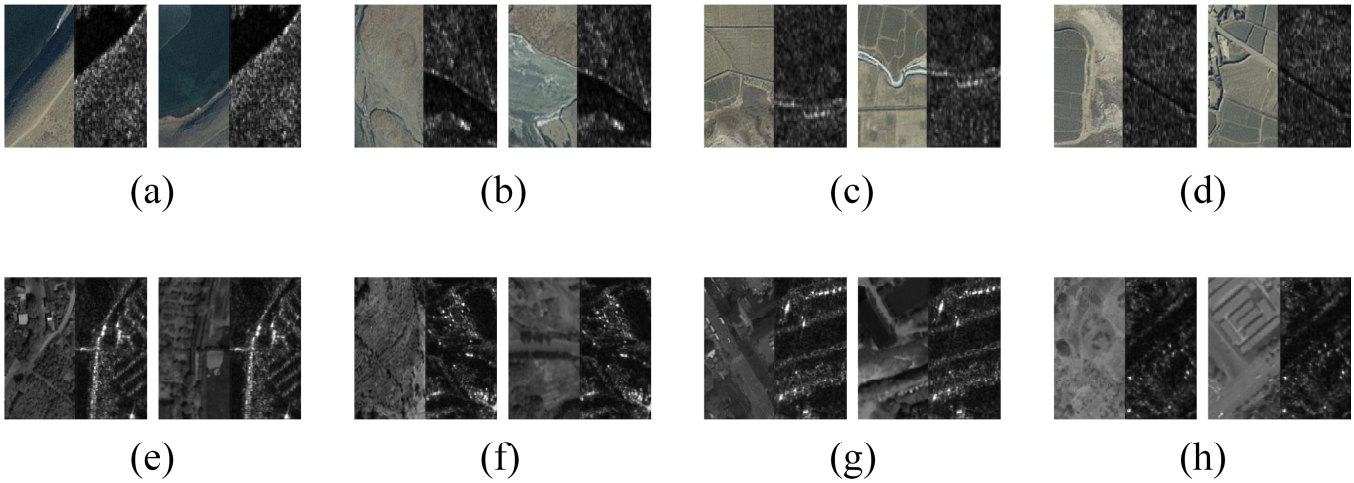

Fig. 28. Comparison of the red inspection area in Fig. 27 before and after registration. (a) Region a in Fig. 27(a). (b) Region b in Fig. 27(a). (c) Region c in Fig. 27(a). (d) Region d in Fig. 27(a). (e) Region e in Fig. 27(b). (f) Region f in Fig. 27(b). (g) Region g in Fig. 27(b). (h) Region h in Fig. 27(b).

points from each image set and assessed the registration accuracy through manual measurements. Additionally, we selected SFOC, one of the most advanced area-based methods, for comparative analysis. The SFOC-based image registration program was obtained from the official website released by its authors.

Table VI presents the matching results of the two methods. Notably, SFOC failed to match the data from Dujiangyan City. Further investigation revealed that the relative positioning error between the original optical and SAR images exceeded 800 m, while the maximum search window in the registration program provided by the SFOC authors was limited to 500 pixels, which was the primary cause of the failure. Therefore, we performed manual coarse registration on the Dujiangyan City dataset to reduce its relative positioning error (Dujiangyan City*). It can be observed that under successful matching, the RMSE of ECSS is comparable to that of SFOC. Generally, area-based methods can achieve higher accuracy than feature-

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based methods, but feature-based methods have the advantage of handling larger relative errors. By introducing a coarse-to-fine matching process, ECSS achieves accuracy comparable to area-based methods. Additionally, area-based methods heavily rely on prior information about the images and can only handle images with translational differences, failing to process images with complex rotational and scale differences, which limits their applicability.

V. Conclusion

In this article, we introduce an advanced feature matching algorithm named ECSS, capable of effectively matching multimodal images with significant scale and rotational differences. ECSS constructs scale space through edge consistency filtering to preserve the structural information of images. To fully harness the information from multiorientation filters, we introduced the significant relative response combined with AB-SIFT descriptors to create feature description vectors that effectively resist NRD. Additionally, to achieve rotational invariance, we employed a stable and fast method for primary orientation estimation. We compared ECSS with nine existing advanced methods (ReDFeat, MINIMA-LG, RIFT, MS-HLMO, SRIF, WSSF, POS-GIFT, OFM, GLS-MIFT), and the results demonstrate that ECSS outperforms these methods on all metrics. Furthermore, ablation experiments conducted by replacing key techniques in ECSS with existing methods further validate the advancement of these technologies. Finally, we validated the engineering applicability of the proposed algorithm on a real earthquake dataset and conducted a comparative analysis with SFOC, one of the advanced area-based methods.

Despite the good performance of the ECSS algorithm in experiments, it still has some limitations. First, ECSS does not possess affine invariance, making it difficult to achieve correct matches in images with large skew angles. Second, the matching performance of ECSS is greatly influenced by the structural features of the images, which leads to poor matching results in images with less distinct structures. Additionally, to more accurately locate feature points, we computed multiple layers in each octave when constructing the scale space, which increased the time cost of the algorithm. Therefore, future research will aim to develop descriptors with affine invariance and enhance the algorithm's performance on images with weaker structural features. This includes refining both the precision and speed of the matching process to achieve better results.

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